

1. Consider the function $f(x) = \sqrt{2x+10}$.(a) Use the Formal Definition of Derivative at a Point (FDoDaP) to determine the value of $f'(3)$.

Use proper limit notation and clearly show all your algebraic steps. [3 pts]

$$\lim_{x \rightarrow 3} \frac{\sqrt{2x+10} - \sqrt{16}}{x-3} = \lim_{x \rightarrow 3} \frac{\sqrt{2x+10} - 4}{x-3} \left(\frac{\sqrt{2x+10} + 4}{\sqrt{2x+10} + 4} \right) = \lim_{x \rightarrow 3} \left(\frac{2x+10-16}{(x-3)(\sqrt{2x+10}+4)} \right)$$

$$= \lim_{x \rightarrow 3} \left(\frac{2(x-3)}{(x-3)(\sqrt{2x+10}+4)} \right) = \lim_{x \rightarrow 3} \left(\frac{2}{\sqrt{2x+10}+4} \right) = \frac{2}{\sqrt{16}+4} = \frac{2}{4+4} = \frac{2}{8} = \boxed{\frac{1}{4}} \quad \checkmark$$

(b) Use the Chain Rule to determine $f'(x)$. Then, evaluate $f'(3)$. [2 pts]

$$f'(x) = \frac{1}{2} (2x+10)^{-\frac{1}{2}} \quad \checkmark$$

$$f'(3) = \frac{1}{4} \quad \checkmark$$

Pro tip: Your answers for $f'(3)$ in parts (a) and (b) should match! $\therefore$ 2. Consider the function $h(x) = 2x^{67}$.(a) Use the Formal Definition of Derivative (FDoD) to determine the derivative $h'(x)$ as a function. Use proper limit notation and clearly show all your algebraic steps. Since your work will involve many terms, you **ARE** allowed to use three dots (...) to express that a pattern will continue. For clarity, list at least three terms before your three dots AND the also list the last term of the pattern. [3 pts]

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{2(x+h)^{67} - 2x^{67}}{h} = \lim_{h \rightarrow 0} \frac{2((x+h)^{67} - x^{67})}{h} = \lim_{h \rightarrow 0} 2 \left(\frac{x^{67} + h \binom{67}{1} x^{66} + h^2 \binom{67}{2} x^{65} + \dots + h^{67} \binom{67}{67} x^0}{h} \right)$$

$$= \lim_{h \rightarrow 0} 2 \left(\binom{67}{1} x^{66} + h \binom{67}{2} x^{65} + h^2 \binom{67}{3} x^{64} + \dots + h^{66} \binom{67}{67} x^0 \right) = \binom{67}{1} x^{66} (2) = 134 x^{66} \quad \checkmark$$

(b) Use the Power Rule to determine $h'(x)$. [1 pt]

$$h'(x) = 134 x^{66} \quad \checkmark$$

Pro tip: Your answers for $h'(x)$ in parts (a) and (b) should match!3. Find the derivative of the following function using any method. Leave your answer **unsimplified**.

$$y = \sqrt[3]{x^2+1} + e^{67} + \sin^5(x^{67}-1) + \sqrt{\pi} \quad (x^2+1)^{\frac{1}{3}} \quad (\sin(x^{67}-1))^5$$

$$y' = \frac{1}{3} (x^2+1)^{-\frac{2}{3}} (2x) + 5 \sin^4(x^{67}-1) \cdot \cos(x^{67}-1) \cdot (67x^{66}) \quad [3 \text{ pts}]$$

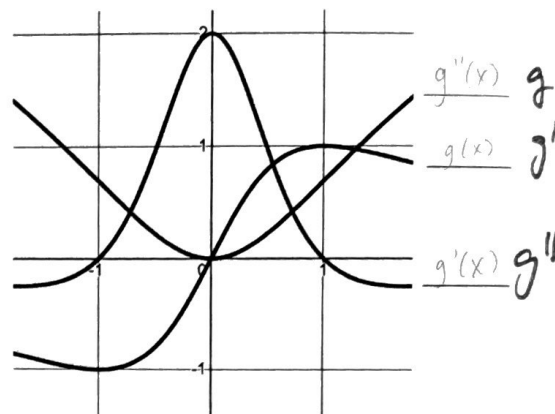
4. Find the equation of the tangent line to the curve $y = 3\cos(5x)$ at the point where $x = \frac{\pi}{3}$. Your answer should NOT include sin or cos. [3 pts] $y' = 3(-\sin(5x)) \cdot 5$

pt = $3\cos\left(5\left(\frac{\pi}{3}\right)\right) = 3\left(-\frac{1}{2}\right) = -\frac{3}{2}$

at $x = \frac{\pi}{3}$, slope = $3(-\sin(\frac{5\pi}{3})) = 3 \cdot \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$

Equation of tangent line: $y - \frac{-3}{2} = \frac{3\sqrt{3}}{2} (x - \frac{\pi}{3})$

5. The diagram below shows the graphs of a function $g(x)$, its first derivative $g'(x)$, and its second derivative $g''(x)$. Use your understanding of the meaning of derivative to identify which graph is which and **clearly label** each using one of the symbols $g(x)$, $g'(x)$, and $g''(x)$ on the diagram. [3 pts]



6. The graph below represents the velocity, in feet per second, of a particle moving along the x-axis over the time interval $[0, 9]$ seconds. [2 pts each]

- (a) Over what time interval(s) is the particle moving left? Justify your answer.

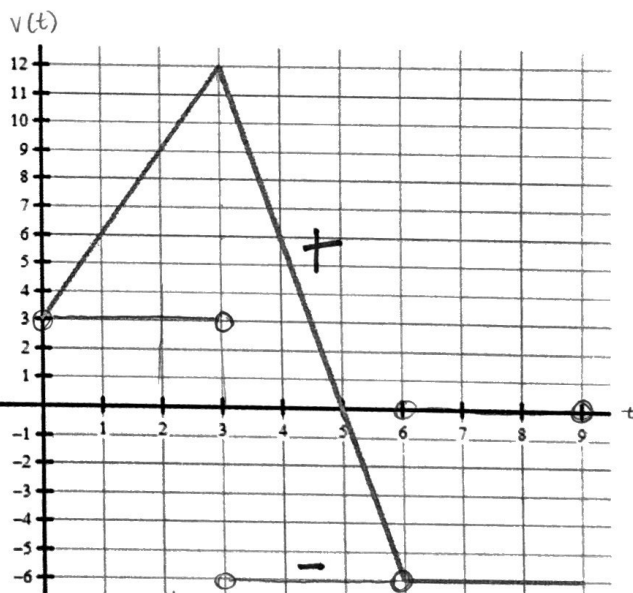
from 5 sec to 9 sec
because velocity is negative
(below x axis)

- (b) At what time t in the given interval is the particle furthest to the right? Justify your answer.

at $t = 5$ because the area is the largest under the curve. The particle has moved to the right as much as possible.

- (c) Over what time interval(s) is the particle speeding up? Justify your answer.

it is speeding up over interval $t = 0$ to $t = 6$ because $v(t)$ and $a(t)$ are either both $(+)$ or both $(-)$



$\leftarrow (0,3), (5,6)$

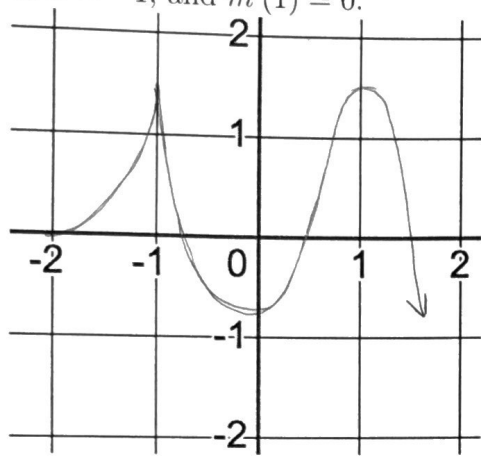
$\frac{3-6}{2} = -\frac{3}{2}$

- (d) What is the average acceleration of the particle over the time interval $[2, 4]$ seconds? Give your answer as a single number with **units**.

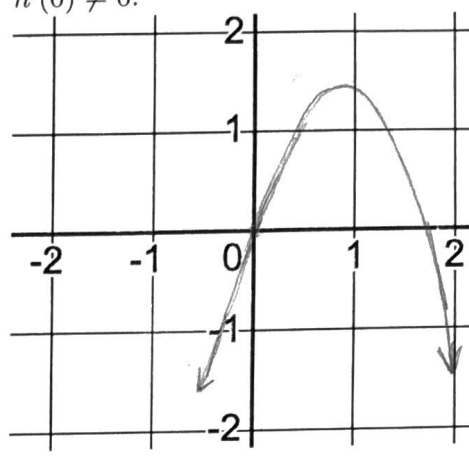
$\frac{-3}{2} \frac{\text{sec}}{\text{ft}^2}$

7. For each of the following, draw a function that fits the description. You are **NOT** required to draw its derivative. If you choose to draw the derivative, make sure to label which is your original function using the symbols $m(x)$ or $n(x)$. [2 pts each]

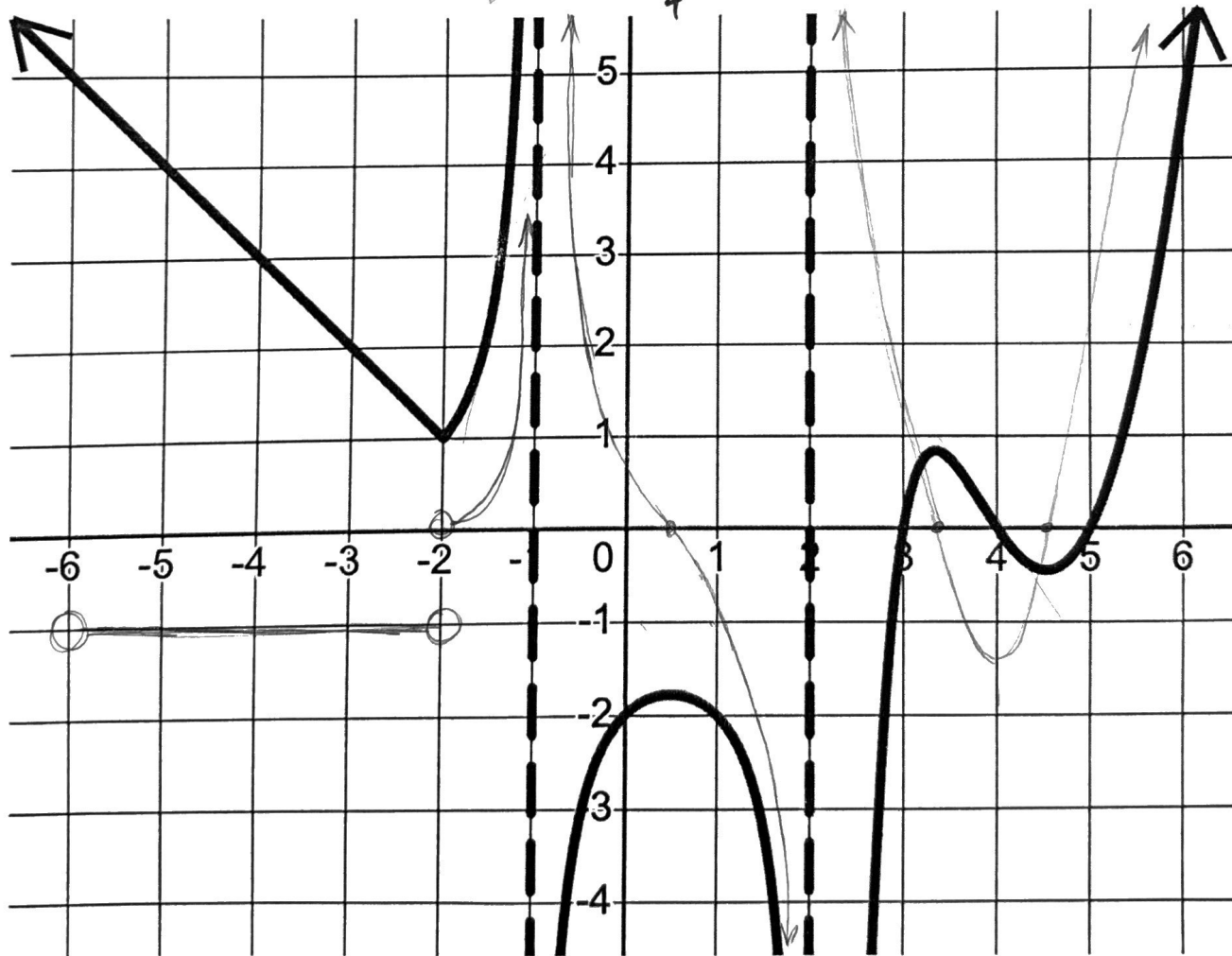
(a) Function $m(x)$ is continuous for all x in the interval $[-2, 2]$, its derivative does not exist at $x = -1$, and $m'(1) = 0$.



(b) Function $n(x)$ whose derivative is a linear function with negative slope, $n(0) = 0$, and $n'(0) \neq 0$.



8. The graph of a function $f(x)$ is given below. Sketch $f'(x)$ in the same diagram. Carefully distinguish full and empty circles in your sketch where relevant. [3 pts]



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